

MACHINE LEARNING

WHILE SO MANY SCI-FI MOVIES HAVE SHOWN HORRIFIC HUMANOID ROBOTS DESTROYING THE HUMAN RACE, WE HAVE CONTINUED TEACHING OUR MACHINES OUR CRITICAL TRICKS. IS THIS THE RIGHT WAY, THOUGH?

BY SOUVIK DAS

Machine learning has been one of mankind's prime points of interest in recent times. Research teams across the world have been building robots, 'teaching' them tasks such as playing football, mixing cocktails and, of late, even giving 'birth'. While research on automation practices are going on, the likes of Stephen Hawking and Elon Musk have expressed their fear of artificial intelligence. How wise is it to allow automated, artificial objects to 'think' on their own?

This brings up a more primitive question: can machines learn to act and behave like humans? A team of scholars at the University of Sussex has stated the improbability of machines ever being able to fully mirror us, and surpass our capability emotion and judgment. After all, without neural qualities such as empathy, collaboration and intuition, how adept would machines be in processing complicated concepts of trust, faith and understanding?

Imparting intelligence into machines are done by programming them to carry out a certain task. For instance, robots mixing cocktails are programmed to visually identify individual bottles by algorithms written by us, and this is taken further by even more code for the robot to measure the amount of gin and vermouth that will be contained in a dry martini recipe. If you own such a robot, it can make you drinks as you enter home after a tiring day behind the desk. But, what if the drink fails to retain the touch that it should? Robots cannot

taste it and see if the drink is the way it should be.

SHAKEN, NOT STIRRED

Till a while ago, machine learning and automation techniques were imparting representational development skills to machines. Artificial intelligence was growing, with features such as facial recognition finding increasingly commonplace application. Shakey the Robot, built by Stanford University's Artificial Intelligence Research wing more than four decades ago, was the first programmed robot that could analyse its own actions to decide the next step. While other machines required fresh set of instructions after completing a large, complicated task, Shakey had the intelligence to create its further set of instructions.

Looking at a specific case in growth of humanoid robots, Honda's development of ASIMO makes for a perfect case study. The affable robot built to assist humans has itself been designed to evolve over time. Its origin goes back to the 1980s, when Honda created its first experimental humanoid, the Honda EO. In the mid-90s, the humanoid evolved into self-controlling robots capable of wireless movement, under Honda's P series. Finally, the entire research data was collated in ASIMO, who premiered in October 2000. Since then, ASIMO has learnt how to play football, offer drinks, climb staircases, and lately, run at a top speed of 3.7kmph. ASIMO can also sense the presence of a human in its path, and step aside to let him pass.

Taking evolution further, a team led by Luiz Brodbeck at the University of



An accident involving such industrial robots claimed a life earlier this year